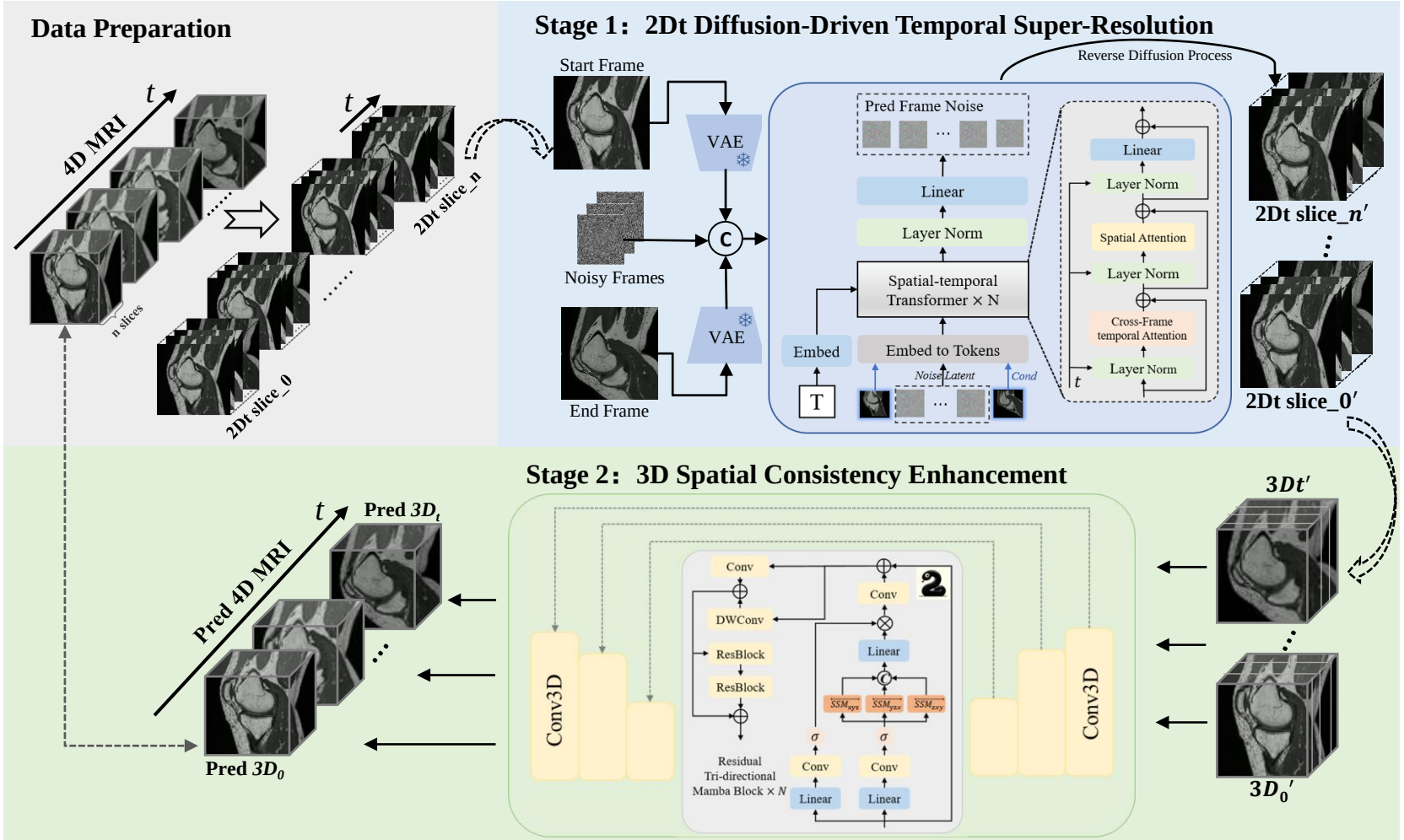
Data Preparation
4D MRI
t
n slices
2Dt slice_0
2Dt slice_n
Stage 1: 2Dt Diffusion-Driven Temporal Super-Resolution
Start Frame
VAE
Noisy Frames
End Frame
C
Reverse Diffusion Process
Pred Frame Noise
Linear
Layer Norm
Spatial-temporal Transformer × N
Embed
Embed to Tokens
Noise Latent
Cond
T
Spatial Attention
Cross-Frame temporal Attention
2Dt slice_n′
2Dt slice_0′
Stage 2: 3D Spatial Consistency Enhancement
Pred 4D MRI
Pred 3D_t
Pred 3D_0
Conv3D
Conv
DWConv
ResBlock
Residual Tri-directional Mamba Block × N
3Dt′
3D_0′